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DBv2: an improved climate-centric calibration-free model for runoff-generation simulation

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ABSTRACT

The calibration-free Dynamic Budyko (DB) model was proposed to address the challenge of streamflow prediction in ungauged basins. In this study, we propose an improved DB model (DBv2), which performed significantly better than the original model (DBv1) with median NSE improving from 0.39 to 0.50 when evaluated across 975 US basins. We then compared DBv2 with two established hydrological models, HBV and GR4J, for runoff generation simulation, allowing all three models to use the same routing module ($DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$; the subscript LR refers to the linear routing structure). The $DBv2_{LR}$, which is calibrated only for runoff-routing, performs as well as fully calibrated HBV_{LR} and $GR4J_{LR}$, with median NSE values being close to 0.65 for all three models. Our study is thus the latest to suggest that a paradigm shift is needed to advance rainfall-runoff modelling by focusing on developing climate-centric hydrological models requiring little or no calibration.

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1 Introduction

The accurate estimate of streamflow is essential in several water resources management applications (e.g. Chiew *et al.* 2003, Archfield *et al.* 2013, Berghuijs *et al.* 2014, Krajewski *et al.* 2017, Rad *et al.* 2017, Singh and Biswal 2019, He *et al.* 2022, Kumar *et al.* 2022, Chembolu and Dutta 2024). Developing a simple, robust and calibration-free hydrological model has been a central interest for hydrologists (e.g. Jakeman and Hornberger 1993, Blöschl and Sivapalan 1995, T. Wagener *et al.* 2001, Krajewski *et al.* 2017, Zheng *et al.* 2018, Gao *et al.* 2019, Merz *et al.* 2022). However, there is no universal set of equations (a universal model) that explains the transformation of rainfall into runoff due to complex and interconnected hydrological processes. Runoff generation is an extremely non-linear and spatially varying process (e.g. Betson 1964, Woods and Sivapalan 1999, Wang *et al.* 2006, Botter *et al.* 2009, Tarasova *et al.* 2020).

In 1969, for the first time, Freeze and Harlan laid (1969) down the outline of the physics-based distributed hydrological model to account for spatial heterogeneity in a basin. The distributed hydrological model applies classical hydrodynamic equations of conservation of mass, energy and momentum at a sub-grid level (Beven 2001, Kampf and Burges 2007, Moradkhani and Sorooshian 2008, Lü *et al.* 2013). In the early 1990s, distributed hydrological models received a lot of attention due to increased computational power and the availability of remote sensing data (e.g. De Troch *et al.* 1990, Giacomelli *et al.* 1995, Kite and Pietroniro 1996, Sandholt *et al.* 1999). Ideally, distributed physics-based hydrological models should outperform their lumped version that approximates the processes at a basin scale. However, in reality, studies found mixed results (e.g. Beven 1989, Grayson *et al.* 1992). Few studies reported that the lumped model outperform distributed models (Reed *et al.* 2004). Moreover, it is not well defined how much fine discretization of process represents reality that leads to better prediction. Also, detailed process representation often leads to a more complex model structure with more parameters, demanding larger datasets for calibration. Due to these limitations, but not limited to, the use of distributed physics-based models was limited in practical applications (Reed *et al.* 2004).

On the other hand, conceptual models demand much less data and computational resources while giving similar performance (Bergstrom 1992, Perrin *et al.* 2003, Moore 2007). These models focus on conceptualizing the dominant hydrological processes, which is why they typically do not employ a large number of free parameters (Bergstrom 1992, Perrin *et al.* 2003, Wagener *et al.* 2004, Moore 2007, Biswal 2016). A typical conceptual hydrological model consists of two major components, soil moisture accounting (SMA) and runoff-routing. The SMA module partitions rainfall into different fluxes such as evapotranspiration, groundwater, runoff, etc. (e.g. Lindström *et al.* 1997, Perrin *et al.* 2003, Moore 2007, P. C. Nayak *et al.* 2013, Manoj *et al.* 2022). The SMA typically follows an infiltration excess (e.g. Liang *et al.* 1994, Soulis *et al.* 2009) or saturation excess (e.g. Bergstrom 1992, Perrin *et al.* 2003) mechanism for runoff generation, followed by a routing module to obtain the streamflow at the basin outlet. On low-permeability soils, when rainfall intensity exceeds the infiltration rate, surface runoff is generated following the infiltration excess mechanism. On the other hand, in high-permeability

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soils, the water table in saturated areas rises, exceeding storage capacity and leading to lateral flow following the saturation excess mechanism (Wood *et al.* 1990).

However, it is possible that different parts of the catchment may follow different runoff generation mechanisms at the same time or at different times due to the spatial heterogeneity of soil (Beven 1983, Reed *et al.* 2004, Sarkar *et al.* 2015). So, in SMA, the soil heterogeneity is accounted for using free parameters, and these parameters are obtained by model calibration using observed streamflow time series (R. Singh *et al.* 2014). The basic assumption here is that climate and catchment characteristics (more specifically soil properties) govern the runoff generation process. So, almost all hydrological model frameworks are soil-centred (Gao *et al.* 2023). However, recent studies have demonstrated that for natural basins, runoff generation is largely controlled by climate (Biswal 2016, de Boer-euser *et al.* 2016, de Boer-Euser *et al.* 2019, Gao *et al.* 2019). This helps to exclude the explicit consideration of soil properties and focus on only climate to predict runoff (except in specific types of basins such as karst landscapes or frozen soil regions).

A convincing argument supporting the exclusion of explicit representation of soil properties in a hydrological model is provided by the Budyko model, which performs hydrological partitioning using only climatic inputs (Budyko 1948, 1974). However, the applicability of the Budyko framework is limited to large timescales, preferably greater than one year (Budyko 1948, Sankarasubramanian and Vogel 2003, 2004, Berghuijs *et al.* 2014, Meira Neto *et al.* 2020, Vora and Singh 2021). Nevertheless, the large timescale hydrological partitioning is ultimately the result of short timescale partitioning. In other words, climate inputs should also govern hydrological partitioning at small timescales. A dynamic hydrological model based on the Budyko curve was first developed by Biswal (2016); it employs a decay function to define the instantaneous dryness index. It has been evaluated in a large number of basins in the USA and India (Biswal 2016, Nag and Biswal 2019, Patnaik *et al.* 2019), and its performance appears reasonable compared to established hydrological models with regionalized parameters (Istalkar and Biswal 2024). However, the original dynamic Budyko (DB) model (referred to here as DBv1) has not undergone any modification, and it may not be capable of fully exploiting climatic information for predicting discharge. In the current study, we propose an improved DB model (DBv2) and then compare it with HBV and GR4J, particularly for modelling flow generation at a daily timescale.

2 Data and methodology

2.1 Data

This study used publicly available datasets, Catchment Attributes and Meteorology for Large-Sample Studies (CAMELS) (Addor *et al.* 2017, Newman *et al.* 2014, 2015) and Model Parameter Estimation Experiment (MOPEX) (Duan *et al.* 2006, Schaake *et al.* 2006), for analysis. These datasets cover basins from a wide range of hydro-climatic regions situated in the Contiguous United States (CONUS). The CAMELS dataset consists of 671 basins with long gauge records (1980–2014) and areas ranging from 4 to 25 000 km^2. We used daily precipitation (mm/day), minimum air temperature (°C), maximum air temperature (°C) and observed streamflow (mm/day) for 607 basins (out of 671), as the other 64 basins have discrepancies in area calculation (Newman *et al.* 2017). The potential evapotranspiration (mm/day) is estimated using the Hargreaves method (Hargreaves and Samani 1985), which requires daily minimum and maximum temperature and latitude.

The MOPEX dataset consists of 416 basins, and we found 48 common basins with CAMELS dataset. So, we obtained the daily precipitation (mm/day), potential evapotranspiration (mm/day), daily air temperature (°C) and observed daily streamflow (mm/day) for 368 MOPEX basins (out of 416) from ftp://hydrology.nws.noaa.gov. The selected basins have data available for periods ranging from 10 to 55 years between the years 1948 and 2002.

In total, this study used data from 975 (607 CAMELS + 368 MOPEX) basins in the CONUS. The long-term data of climate, soils, topography and vegetation for these basins is obtained from Falcone *et al.* (2010).

2.2 Modified dynamic Budyko model

At a large timescale (greater than a year), rainfall partitioning into evapotranspiration (ET) and streamflow (Q) can be given by the well-known Budyko framework (Budyko 1948). The framework is based on the dryness index ($\emptyset$) (also called the aridity index), which is a ratio of long-term mean potential evapotranspiration (PET) and rainfall (R). According to this framework,

$$< Q > = < R > \cdot f(\emptyset) \tag{1}$$

$$f(\emptyset) = 1 - \left(\emptyset \cdot \tanh\left(\emptyset^{-1}\right)\left(1 - e^{-\emptyset}\right)\right)^{0.5} \tag{2}$$

where $\langle Q \rangle$, $\langle R \rangle$ and $f(\emptyset)$ are the long-term mean streamflow, long-term mean rainfall and Budyko function, respectively. However, this framework cannot be used for continuous streamflow modelling, and is valid only for large time scales where the change in storage is negligible in the mass balance equation (Budyko 1948, 1974). Recently, Biswal (2016) proposed a DB model framework using a top-down approach that simulates the dryness index at a daily timescale and ultimately streamflow using the Budyko framework.

The DB model framework divides the river basin into two hypothetical zones; zone I is responsible for ET generation, and zone II generates ET and streamflow (see Fig. 1 for reference). This assumption is quite close to reality as very low-intensity rainfall does not produce streamflow. In zone I, if R > PET, then after satisfying potential evapotranspiration demand, the remaining rainfall water (W) goes to zone II. Similarly, if PET > R, leftover solar energy (H) goes into zone II. Mathematically,

$$W(t) = R(t) - PET(t); if\ R(t) > PET(t),\ else\ 0 \tag{3}$$

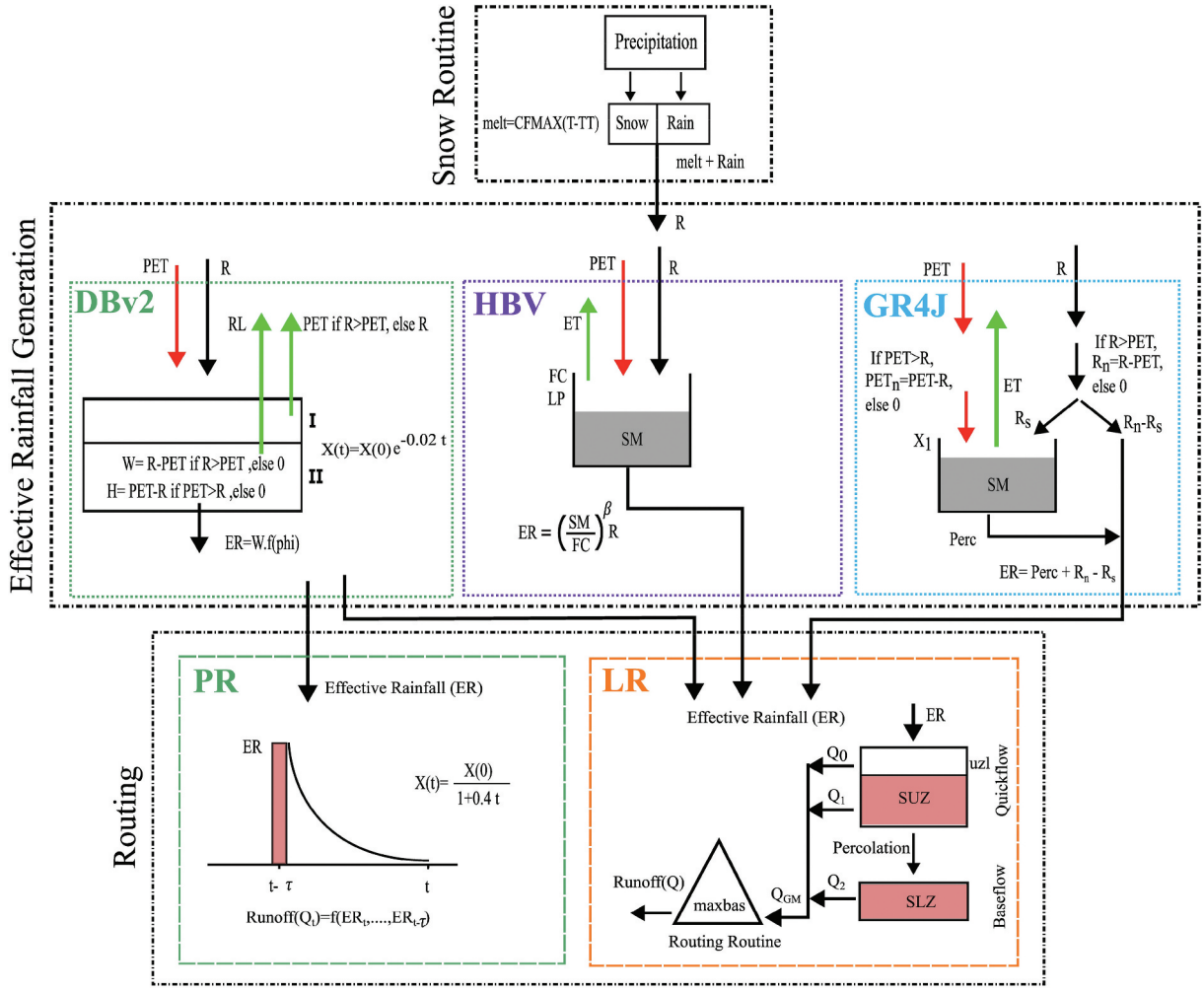


Figure 1. Graphical illustration of $DBv2_{PR}$, $DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$ model structure. The snow routine module is used to estimate snowmelt with the degree-day method. The $DBv2_{PR}$ model uses an exponential decay function for effective rainfall generation and a power law-based decay function (PR) in routing to simulate streamflow at the basin outlet. For the $DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$ models, effective rainfall generated using DBv2, HBV and GR4J runoff generation modules, respectively, is transformed into streamflow using the linear reservoirs routing (LR) module.

$$H(t) = PET(t) - R(t); if\ PET(t) > R(t), else\ 0 \quad (4)$$

In zone II, if no ET demand is present, all water (W) transforms into streamflow. However, in the case of ET demand, the interaction of W and H decides the partitioning of W into rainfall loss (RL) and effective rainfall (ER). The fraction of W that becomes evapotranspiration is called RL, while the fraction of W that remains after meeting evapotranspiration (ET) demand and transforms into streamflow is called ER. Similar to long-term hydrological partitioning using the Budyko framework (see Eq. 1), ER at a daily timescale is given as

$$ER(t) = W(t) \cdot f(\varphi(t)) \quad (5)$$

In the above equation φ is the instantaneous dryness index (Biswal 2016) and $f(\varphi)$ is the Budyko function (Eq. 2). The value of φ at time t is estimated considering the antecedent effect of W and H Again, similar to the long-term dryness index of the Budyko framework, φ is estimated as

$$\varphi(t) = \frac{FH(t)}{FW(t)} \quad (6)$$

The functions $FW(t) = \int_{t-N}^{t} x(t-\tau)W(\tau)d\tau$ and $FH(t) = \int_{t-N}^{t} x(t-\tau)H(\tau)d\tau$ represent the available water and available energy, respectively, at time t, where τ is a dummy variable and N is number of past days. The function $x(t)$ is the decay function, giving us the value of x remaining at time t from the original input $x(0)$ (x at time $t = 0$), which was found to be a power-law function:

$$x(t) = x(0)(1 + 0.4 \cdot t)^{-1} \quad (7)$$

This decay function was obtained by performing a recession flow analysis using 15 basins randomly selected from the MOPEX dataset (Biswal 2016). Note that the model results are not very sensitive to N, and its value is fixed at 365 to ensure that past inputs are given due importance. Ultimately, ER is transformed into streamflow (Q)

$$Q(t) = \int_0^t -\frac{\partial}{\partial t}(x(t-\tau)ER(\tau)) \cdot d\tau \quad (8)$$

using the same power law-based decay function (Equation 7).

The potential of the DB model has been tested in various applications, including the prediction of daily streamflow (Biswal 2016, Patnaik *et al.* 2019, Istalkar and Biswal 2024), prediction of flow-duration curves in ungauged basins (Nag and Biswal 2019), soil moisture data assimilation (Nayak *et al.* 2021a, 2021b) and prediction of drought hotspot maps (Nayak *et al.* 2022). Considering the fact that DB is calibration-free, its performance is acceptable for prediction in ungauged basins (Istalkar and Biswal 2024). However, the reasons behind DB's lower performance in comparison to a fully calibrated hydrological model remain unexplored (Istalkar and Biswal 2024). Our main aim here is to understand whether the runoff generation module or the runoff routing module needs improvement. To address this question, we focus on improving the runoff generation module. The core idea in DB involves the estimation of the instantaneous dryness index (φ), which controls the effective rainfall generation and, ultimately, overall model performance. The estimation of φ is mainly governed by the power-law based decay function (Equation 7), derived from recession flow analysis (Biswal 2016). In this study, we plan to test the exponential decay function, which is also commonly used (Brutsaert 2008, 2010):

$$x(t) = x(0) \cdot e^{-a \cdot t} \tag{9}$$

In this equation, a is the decay coefficient (per unit time), $x(0)$ represents the initial quantity, and $x(t)$ denotes the quantity remaining at time t. The study by Brutsaert (2010) reported that the value of a remains fairly constant across river basins and is typically 0.02 (1/50 day^{-1}). Thus, in this study, we selected the value of a as 0.02 (day^{-1}). Since the value of the parameter is determined from the existing literature, the runoff-generation module can be called calibration-free. The *FW* and *FH* based on the proposed exponential decay function are given as

$$FW(t) = \int_0^t W(\tau) \cdot e^{-0.02 \cdot (t-\tau)}\, d\tau \tag{10}$$

$$FH(t) = \int_0^t H(\tau) \cdot e^{-0.02 \cdot (t-\tau)}\, d\tau \tag{11}$$

Ultimately, the instantaneous dryness index, effective rainfall, and streamflow at the basin outlet are estimated similarly to Equations (5), (6) and (8). The original DB model (DBv1) proposed by Biswal (2016) simulates effective rainfall and streamflow at the basin outlet using a power law-based (PR) decay function (termed DBv1$_{PR}$ hereafter). The modified DB model (DBv2) uses an exponential decay function in effective rainfall simulation and a power law-based (PR) decay function in predicting streamflow at the basin outlet (termed DBv2$_{PR}$ hereafter; see Fig. 1). Moreover, the effective rainfall simulated by DBv2 model is fed to two linear reservoirs connected parallel to predict streamflow at the basin outlet, as discussed in detail in section 2.5.

2.3 HBV runoff generation model

HBV (Hydrologiska Byråns Vattenbalansavdelning) is a widely used conceptual rainfall-runoff model developed by the Swedish Meteorological and Hydrological Institute (Bergstrom 1992). The successful application of HBV in a wide range of climatic regions and several studies (e.g. Lindström *et al.* 1997, Kobold and Brilly 2006, Singh and Bárdossy 2012, Beck *et al.* 2016, Seibert and Bergström 2022, Istalkar and Biswal 2024), and its computational efficiency and flexibility (e.g. Dakhlaoui *et al.* 2012, Hailegeorgis and Alfredsen 2015, Montero *et al.* 2016, Kadu and Biswal 2022) are a few reasons for selecting it in the current analysis. The HBV model consists of a soil moisture routine, response routine and routing routine. The soil moisture routine is responsible for runoff generation and accounts for the dynamics of soil moisture using two free parameters, *FC* and β. The fraction of rain (R) that transforms into effective rainfall (ER) is given as

$$ER = \left(\frac{SM}{FC}\right)^{\beta} \cdot R \tag{12}$$

where *FC* (mm) and *SM* (mm) are the maximum and actual water storage in the unsaturated zone. β is the shape coefficient of the recharge function. The rainfall loss, i.e. actual evapotranspiration (*ET*) from the soil moisture routine, is estimated using the available soil moisture *SM* and free parameter *LP*.

$$ET = PET \cdot min\left(\frac{SM}{FC \cdot LP}, 1\right) \tag{13}$$

The estimated *ET* is used to update the *SM*. Ultimately, the streamflow at the basin outlet is obtained by feeding ER to the routing model discussed in section 2.5 (see Fig. 1).

2.4 GR4J runoff generation model

GR4J is a conceptual lumped rainfall-runoff model with four freeparameters (Perrin *et al.* 2003) that is used in a wide range of water resources applications (e.g. Humphrey *et al.* 2016, Stephens *et al.* 2018, Sezen *et al.* 2019). It is a modified version of the GR3J model originally proposed by Edijatno and Michel (1989) and updated over successive years by Nascimento (1995) and Edijatno *et al.* (1999). GR4J shows significant improvement over GR3J in simulating low flows (Perrin *et al.* 2003). This model, similar to HBV, belongs to the group of soil moisture accounting models. GR4J model formulation first estimates the net rainfall (R_n) and net potential evapotranspiration (PET_n) using the equations below.

$$R_n = R - PET \ \ if \ \ R > PET, else \ 0 \tag{14}$$

$$PET_n = PET - R \ \ if \ \ PET > R, else \ 0 \tag{15}$$

The estimation of R_n and PET_n is very similar to that of water (W) and solar energy (H) reaching zone II after accounting for the losses in zone I of DB model (see Fig. 1). Then, the portion of R_n, R_s goes into the production store where percolation and actual evapotranspiration (ET) are computed based on the available soil moisture (SM) and a free parameter X_1.

$$R_s = \frac{X_1\left(1-\left(\frac{SM}{X_1}\right)^2\right)tanh\left(\frac{R_n}{X_1}\right)}{1+\frac{SM}{X_1}tanh\left(\frac{R_n}{X_1}\right)} \quad (16)$$

$$ET = \frac{SM\left(2-\frac{SM}{X_1}\right)tanh\left(\frac{PET_n}{X_1}\right)}{1+\left(1-\frac{SM}{X_1}\right)tanh\left(\frac{PET_n}{X_1}\right)} \quad (17)$$

The soil moisture in the production store is updated after considering the ET losses. The percolation $Perc$ is computed as

$$Perc = SM\left\{1-\left[1+\left(\frac{4}{9}\frac{SM}{X_1}\right)^4\right]^{-1/4}\right\} \quad (18)$$

Finally, the effective rainfall (ER) is computed considering all losses; $ER = Perc + R_n - R_s$. This ER is ultimately transformed into streamflow using the routing model discussed in section 2.5 (see Fig. 1).

2.5 Effective rainfall to streamflow: routing

The effective rainfall simulated by the DBv2, HBV and GR4J runoff generation models is fed to a routing module to obtain the streamflow at basin outlet. To compare the robustness of the proposed DBv2 runoff generation model with HBV and GR4J runoff generation models, we use a similar routing model for all these runoff generation models to obtain the streamflow at the basin outlet. We employ the HBV model's own routing structure having linear reservoirs (LR) connected in parallel (Seibert and Vis 2012) (see Fig. 1). The routing module is comprised of two reservoirs: storage in the upper zone (SUZ) and storage in the lower zone (SLZ). These two reservoirs denote the two soil zones from which quick flow and baseflow are generated. The ER enters the upper reservoir and leaves through three paths: (1) Q_0, outflow from SUZ, controlled by recession coefficient K_0, when SUZ > UZL (threshold parameter for quick flow); (2) quick flow, Q_1, outflow from SUZ, controlled by recession coefficient K_1 and; (3) percolation to SLZ with constant percolation rate (PERC). Finally, Q_2, the baseflow from SLZ, is computed using the recession coefficient K_2. Ultimately, the outflow from reservoirs (Q_{GM}) is routed using the triangular weighting function MAXBAS to get the streamflow at basin outlet (Q).

$$Q_{GM} = K_0 \cdot (SUZ - UZL) + K_1 \cdot SUZ + K_2 \cdot SLZ \quad (19)$$

$$Q = \sum_{i=1}^{MAXBAS} c(i) Q_{GM}(t-i+1) \quad (20)$$

where $c(i) = \int_{i-1}^{i} \frac{2}{MAXBAS} - \left|u - \frac{MAXBAS}{2}\right| \frac{4}{MAXBAS^2} du$.

The DBv2, HBV and GR4J runoff generation modules coupled with the above linear reservoir routing (LR) module are termed the $DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$ models, respectively (see Fig. 1).

2.6 Model calibration and performance evaluation

The $DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$ models were calibrated using observed streamflow time series, with Nash-Sutcliffe efficiency (NSE) as the objective function (Nash and Sutcliffe 1970) and genetic algorithm (GA) as the optimization algorithm in MATLAB. GA is one of the most commonly used algorithms and has proven to be effective in solving optimization problems (e.g. Bekele and Nicklow 2007, Maier *et al.* 2014, Beck *et al.* 2016, R. K. Singh *et al.* 2023). The criteria used to divide the meteorological inputs and observed streamflow time series into calibration and validation sets are subjective (Beck *et al.* 2016, Gao *et al.* 2019, Mai 2023). We divided the complete time series for each basin into 50% for calibration and 50% for validation, ensuring sufficient data in both cases.

The study basins receive precipitation in the form of snow. We employed a simple degree-day-based snow routine model to estimate snow accumulation and snowmelt. According to this framework, when the temperature (T,°C) is below the threshold temperature (TT, °C), precipitation accumulates as snow (mm/day). The snow correction factor ($SFCF$) is applied to account for undercatch correction. When T rises above TT, snowmelt is given as

$$melt = CFMAX(T - TT) \quad (21)$$

where $melt$ and $CFMAX$ are snowmelt (mm/day) and snowmelt rate (mm °C^{-1} day^{-1}), respectively. All the meltwater does not go to the soil. Some water is held by the snowpack (mm/day), and it is accounted using the free parameter CWH, representing the water holding capacity of snow.

$$water\ held\ by\ snowpack = CWH \cdot Snow \quad (22)$$

When T is below TT, the refreezing of free water (mm/day) in snow is modelled using the refreezing coefficient CFR.

$$refreezing = CFR \cdot CFMAX(TT - T) \quad (23)$$

Finally, the water (R = rain + snowmelt) along with PET is fed into the effective rainfall generation module of $DBv2_{PR}$, $DBv2_{LR}$, HBV_{LR} and $GR4J_{LR}$ (see Fig. 1). As this study focuses on runoff generation, calibrating the snow routine parameters for each model individually will lead to different sets of optimum snow parameters, resulting in different water inputs to the models at a given time t. Thus, to provide consistent water input at time t to all models, we use the snow routine parameters obtained during HBV_{LR} calibration for all models, i.e. $DBv1_{PR}$, $DBv2_{PR}$, $DBv2_{LR}$, and $GR4J_{LR}$. Thus, HBV_{LR} is calibrated for parameters in the snow routine, effective rainfall generation, and runoff routing; $GR4J_{LR}$ for effective rainfall generation and runoff routing; and $DBv2_{LR}$ only for runoff routing. Note that in the case of $DBv1_{PR}$ and $DBv2_{PR}$, we use snow parameters from the calibrated HBV model and still refer to these models as calibration-free. This is because the snow module is not an inherent part of the hydrological model and can be excluded by considering only non-snow-affected basins. However, doing so would significantly reduce the number of basins available for this analysis and limit the generalizability of the study outcomes. The calibrated parameters and their respective ranges are detailed in Table 1.

Table 1. Free parameters along with their ranges for the snow routine model, HBV and GR4J runoff generations models, and linear reservoir routing model.

Model	Parameter	Description	Minimum	Maximum
Snow routine model	TT(°C)	Threshold temperature below which precipitation is assumed to be snow	−2.5	2.5
	SFCF	Snowfall correction factor	1	1.5
	CWH	Water holding capacity of snow	0	0.2
	CFR	Refreezing coefficient	0	0.1
	CFMAX (mm °C^{-1} day^{-1})	Snowmelt rate	0.5	5
HBV runoff generation model	FC (mm)	Maximum water storage in unsaturated zone	50	700
	β	Shape coefficient	1	6
	LP	Threshold at which ET is equal to PET	0.3	1
GR4J runoff generation model	X_1 (mm)	Maximum capacity of production store	1	1200
Linear reservoir routing model	UZL (mm)	Threshold parameter for quick flow	0	100
	PERC (mm day^{-1})	Percolation to lower zone	0	6
	K_0 (day^{-1})	Additional recession coefficient of upper zone	0.05	0.99
	K_1 (day^{-1})	Recession coefficient of upper zone	0.01	0.8
	K_2 (day^{-1})	Recession coefficient of lower zone	0.001	0.15
	MAXBAS (day)	Length of triangular weighting function	1	3

In this study, we evaluate model performance using coefficient of determination (R^2), NSE (Nash and Sutcliffe 1970) and bias (B) metrics (Beck *et al.* 2016).

$$R^2 = \left(\frac{\sum_{i=1}^{n} \left(Q_o^i - \bar{Q}_o\right)\left(Q_m^i - \bar{Q}_m\right)}{\sqrt{\sum_{i=1}^{n} \left(Q_o^i - \bar{Q}_o\right)^2} \sqrt{\sum_{i=1}^{n} \left(Q_m^i - \bar{Q}_m\right)^2}} \right)^2 \quad (24)$$

$$NSE = 1 - \frac{\sum_{i=1}^{N} \left(Q_o^i - Q_m^i\right)^2}{\sum_{i=1}^{N} \left(Q_o^i - \bar{Q}_o\right)^2} \quad (25)$$

$$B = 1 - \left| \frac{\bar{Q}_m - \bar{Q}_o}{\bar{Q}_m + \bar{Q}_o} \right| \quad (26)$$

where Q_o, Q_m, $\bar{Q}_m$ and $\bar{Q}_o$ are daily observed, daily modelled, daily mean modelled and daily mean observed streamflow, respectively. The R^2 value ranges between 0 and 1, representing no agreement and perfect agreement, respectively. The NSE ranges between $-\infty$ and 1, and the desirable result is 1, representing perfect agreement between observed and modelled streamflow. B is a unitless measure of the model's water balance error. For no water balance error, its value is 1; otherwise, it is less than 1. The above performance metrics together provide sufficient information to evaluate the model's performance.

3 Results and discussion

3.1 Model performance comparison

Figure 2 compares the performance of $DBv1_{PR}$ and $DBv2_{PR}$ during the validation period in terms of R^2, NSE and B in predicting daily streamflow for 975 study basins. The R^2 interquartile (25th, 50th and 75th) values for $DBv1_{PR}$ and $DBv2_{PR}$, respectively, are (0.43, 0.54, 0.62) and (0.48, 0.59, 0.66). The NSE interquartile values in the same order are (−0.26, 0.39, 0.55) and (0.23, 0.50, 0.61). The $DBv2_{PR}$ outperforms $DBv1_{PR}$ in terms of R^2, indicating its superiority in capturing observed streamflow dynamics. In terms of NSE, $DBv2_{PR}$ is superior to $DBv1_{PR}$ for all interquartile values. Notably, the 25th percentile NSE improved from −0.26 to 0.23. This indicates that the low-performing basins of $DBv1_{PR}$ show significant improvement upon incorporating the exponential decay function in the φ (instantaneous dryness index) value estimation. Moreover, the modified φ improved not only the $DBv2_{PR}$ model's accuracy in

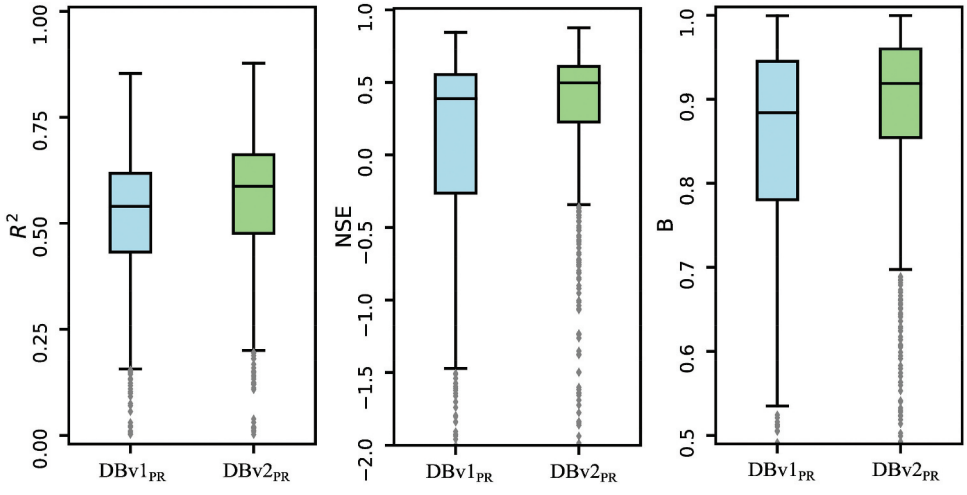


Figure 2. Box plot comparing daily streamflow prediction performance of $DBv1_{PR}$ and $DBv2_{PR}$ models in terms of R^2, NSE and B for 975 study basins. The $DBv1_{PR}$ utilizes a power law-based decay function for effective rainfall simulation and runoff-routing. The DBv2PR uses an exponential decay function for effective rainfall simulation and a power law-based decay function for runoff-routing. Both DBv1PR and DBv2PR models are calibration-free.

predicting daily streamflow but also its water balance. The B interquartile values improved from (0.78, 0.88, 0.95) to (0.85, 0.92, 0.96), indicating the superiority of $DBv2_{PR}$ over $DBv1_{PR}$ in capturing the water balance.

The key difference between power law-based and exponential decay functions, used in DBv1 and DBv2 runoff generation models, respectively, is their decay rate. Figure 3 compares the decay of these two functions for a quantity X (which can be assumed as W or H) with an initial value of 100. From the figure, it is clear that the proposed exponential decay function decays at a slower rate compared to the power law-based decay function. Note that the power-law decay function was chosen because recession flow, which results from the decay of storage, typically follows a power-law profile (e.g. Biswal and Marani 2014, Biswal 2016, Patnaik *et al.* 2018, Sharma and Biswal 2022). Nevertheless, DBv2, with an exponential decay function, provides a superior performance. This contradiction might be explained by the argument that recession flow analysis considers drainable storage, which decays to produce only streamflow (Shaw and Riha 2012, Biswal and Nagesh Kumar 2015, Tashie *et al.* 2020), whereas W represents total storage, which decays to produce both streamflow and evapotranspiration. However, it remains unclear how recession flow analysis based on the exponential decay function assumption (e.g. Brutsaert 2008, 2010) leads us to the correct value of the exponential decay coefficient, $a = 0.02$ per day. Perhaps this is just another example of getting the right answer for the wrong reason (Klemeš 1983).

The value of the exponential decay function coefficient $a = 0.02$ is taken from the literature (Brutsaert 2010) and is assumed to be universal. However, a study by Brutsaert (2010) reported that the value of a can range between 0.01 (day^{-1}) and 0.038 (day^{-1}). Therefore, we analysed the performance of $DBv2_{PR}$ for different a values in terms of NSE and B for given study basins. From Fig. 4, the median NSE values for a of 0.05, 0.01, 0.015, 0.02, 0.025, 0.03 and 0.035 are 0.46, 0.50, 0.51, 0.50, 0.49, 0.47 and 0.46, respectively. The median B values in the same order are 0.88, 0.91, 0.91, 0.92, 0.92, 0.91 and 0.91. In terms of median NSE, the performance of $DBv2_{PR}$ is lowest for a of 0.005, increases with a up to 0.015, and then decreases as a increases up to 0.035. In the case of B, the performance across all B interquartiles (25th, 50th, and 75th) is lowest for a of 0.005, increases with a up to 0.02, and then decreases to 0.035. So, considering median as a representative value, $DBv2_{PR}$ performance is maximum for a of 0.015 for NSE and 0.02 for B. The optimum value of a obtained for the current study basins is quite close to the value suggested in the literature (Brutsaert 2010). Considering that $DBv2_{PR}$ can be an essential tool for streamflow prediction in ungauged basins, where the B metric is important for water resources management, the optimum value of $a = 0.02$ obtained in this study exactly matches the value found in the study by Brutsaert (2010).

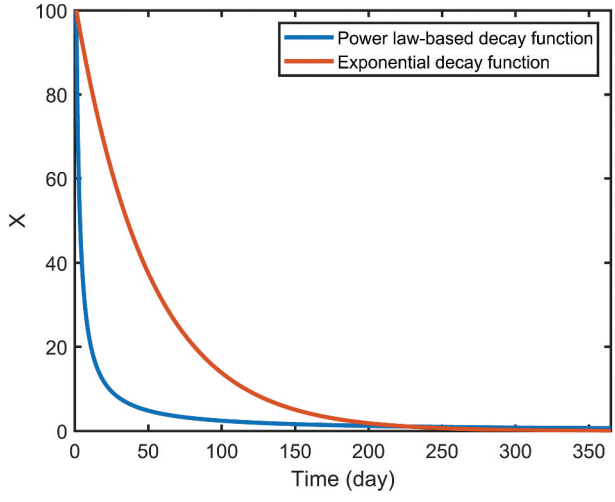


Figure 3. Illustration of how a quantity X (which can be assumed as W or H) decays with time t using a power law-based decay function and the proposed exponential decay function.

Figure 5 compares the performance of $DBv2_{PR}$, $DBv2_{LR}$, $GR4J_{LR}$ and HBV_{LR} models for the validation period in terms of R^2, NSE and B. The R^2 interquartile values for $DBv2_{PR}$, $DBv2_{LR}$, $GR4J_{LR}$ and HBV_{LR}, respectively, are (0.48, 0.59, 0.66), (0.58, 0.68, 0.74), (0.58, 0.68, 0.75) and (0.58, 0.68, 0.75). The NSE interquartile values in the same order are (0.23, 0.50, 0.61), (0.51, 0.64, 0.72), (0.51, 0.65, 0.72) and (0.52, 0.66, 0.73). In terms of R^2 and NSE, the performance of $DBv2_{PR}$ is the lowest when compared to the HBV_{LR} and $GR4J_{LR}$ models. However, considering the fact that $DBv2_{PR}$ is calibration-free, its performance is acceptable (median NSE = 0.50), particularly for streamflow prediction in ungauged basins. The performance of $DBv2_{PR}$ has improved significantly upon considering linear reservoirs as the routing module ($DBv2_{LR}$). Notably, $DBv2_{LR}$ performed equally well to $GR4J_{LR}$ and HBV_{LR} in terms of R^2 (median ≈ 0.68) and NSE (median ≈ 0.65). Note that $DBv2_{LR}$ is calibrated for only the routing module, whereas $GR4J_{LR}$ and HBV_{LR} are fully calibrated. Thus, $DBv2_{PR}$ and $DBv2_{LR}$ have the same runoff-generation module, indicating similar water balance errors with a median B value of 0.92. In the case of $GR4J_{LR}$, the median B value is 0.93, showing slightly better performance compared to $DBv2_{PR}$. However, HBV_{LR} is considerably superior to $DBv2_{LR}$, with a median B of 0.94. The possible explanation behind this is the two extra parameters, i.e. β and LP, in the soil moisture routing of HBV, making it more flexible.

We selected a sample basin #11532500 (41°47′30”N, 124° 04′30”W) having a dryness index of 0.4 and mean daily observed streamflow of 5.54 (mm/day) to investigate the impact of the decay function on the runoff generation and routing module in DBv1 and DBv2. Figure 6 shows the streamflow time series predicted by all models, along with observed streamflow and rainfall for a year. From the figure, it can be observed that $DBv1_{PR}$ (shown in red) tends to overestimate the streamflow peaks (peaks occurring between 1 September and 20 November 1979), when the antecedent soil condition is dry. However, these peaks have decreased significantly in the case of $DBv2_{PR}$ (shown in yellow). The explanation lies in the decay rates of W and H in both models. The power law-based decay function in the runoff generation of DBv1 decays quickly and has a short memory of the past dry conditions of the catchment. This resulted in an underestimation of φ, leading to streamflow overestimation. However, the proposed exponential decay function in the runoff generation of DBv2 decays slowly (see Fig. 4 for reference) and takes into account the past

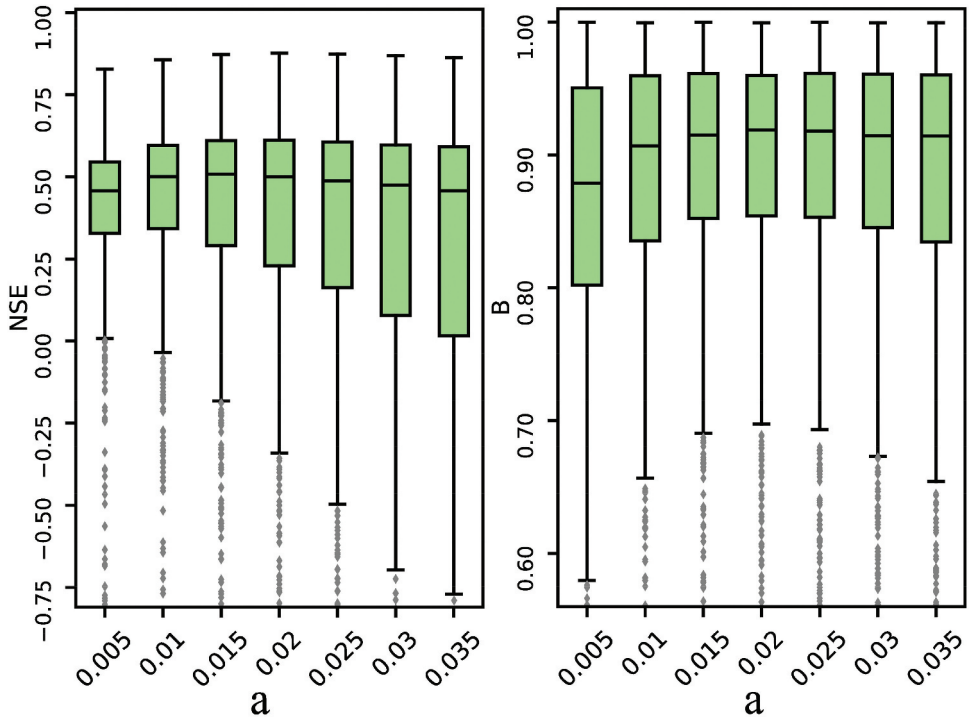


Figure 4. $DBv2_{PR}$ performance for different a values in terms of NSE and B. The optimal performance is achieved at $a = 0.02$ when considering the median B as the representative value.

catchment condition, resulting in an improved estimate of φ and streamflow peaks.

Nevertheless, $DBv2_{PR}$ still overpredicts the peaks (e.g. peak after 28 February 1980) and underpredicts the low flow (e.g. falling limb just before and after 18 April 1980) compared to observed streamflow values. The reason behind this is the lack of separation of quick and slow flow components in $DBv2_{PR}$ routing module, resulting in not so accurate capturing of high and low flow values. However, on replacing $DBv2_{PR}$ model's routing with linear reservoir routing ($DBv2_{LR}$), the performance is as good as that of $GR4J_{LR}$ and HBV_{LR}. There is scope to modify the $DBv2_{PR}$ routing structure by separating the quick and slow flow components; however, this aspect is not considered in this study. Notably, the introduction of $DBv2_{PR}$ attempts to address several long-standing challenges in rainfall-runoff modelling, such as prediction in ungauged basins, non-stationarity of parameters due to changing climate, etc.

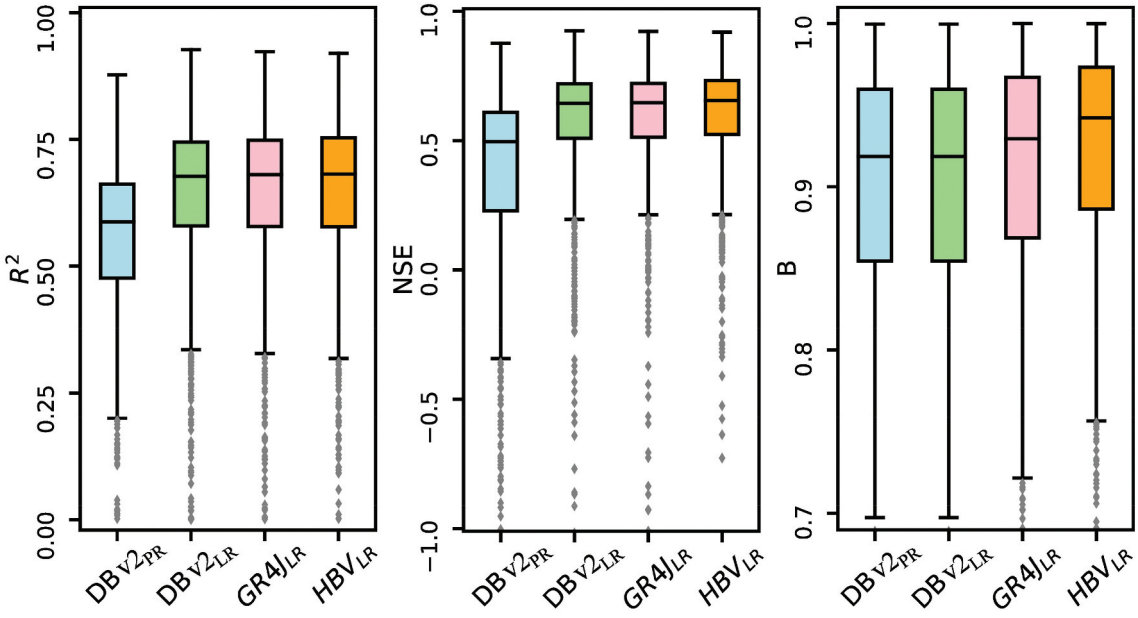


Figure 5. Box plot comparing streamflow prediction performance of $DBv2_{PR}$, $DBv2_{LR}$, $GR4J_{LR}$ and HBV_{LR} in terms of R^2, NSE and B for 975 study basins. $DBv2_{PR}$ is calibration-free and $DBv2_{LR}$ is only calibrated for runoff-routing, whereas $GR4J_{LR}$ and HBV_{LR} are fully calibrated.

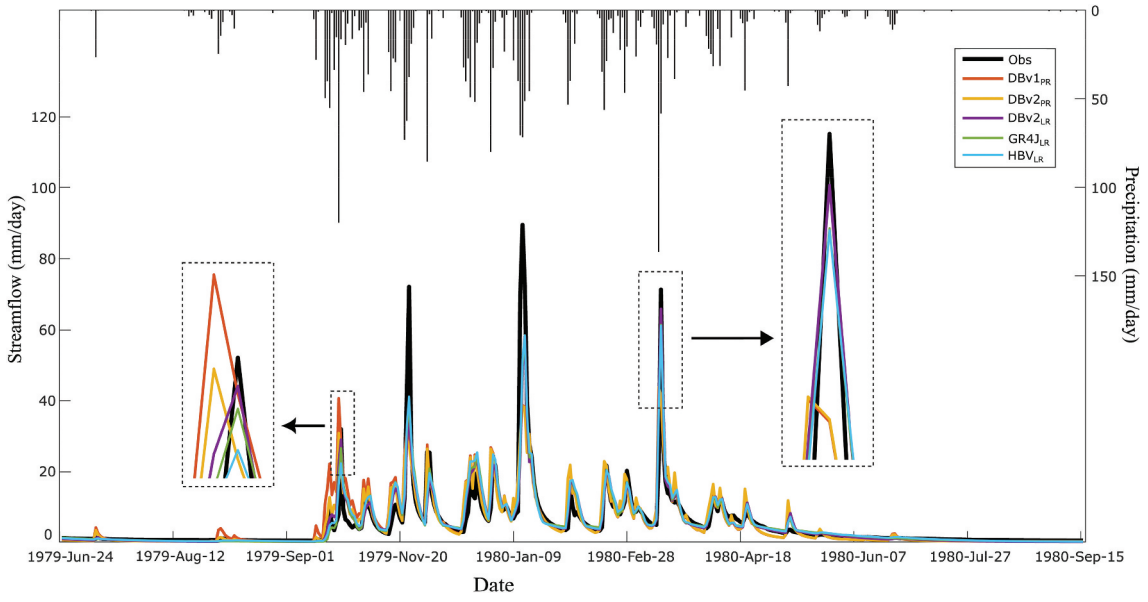


Figure 6. Streamflow time series predicted by $DBv1_{PR}$, $DBv2_{PR}$, $DBv2_{LR}$, $GR4J_{LR}$, and HBV_{LR}, along with observed streamflow and precipitation for sample basin #11532500.

3.2 Factors affecting runoff generation of DBv2

It is desirable to investigate the robustness of the model structure by testing its performance across a wide range of physio-climatic conditions. Therefore, we explored the spatial pattern of $DBv2_{LR}$ performance in terms of NSE by considering HBV_{LR} and $GR4J_{LR}$ as benchmark models. In Fig. 7(b), for basins represented by open grey circles, $DBv2_{LR}$ and HBV_{LR} show similar performance in terms of NSE ($DBv2_{LR}$ – HBV_{LR} = −0.05 to 0.05) in predicting daily streamflow. For basins represented by blue dots, HBV_{LR} outperforms $DBv2_{LR}$ ($DBv2_{LR} - HBV_{LR} < -0.05$), while for green dot basins, $DBv2_{LR}$ is superior to HBV_{LR} ($DBv2_{LR} - HBV_{LR} > 0.05$). For the majority of basins, $DBv2_{LR}$ shows similar performance to HBV_{LR}, but distinct clusters are observed in a few regions where $DBv2_{LR}$ and HBV_{LR} dominate. For instance, in clusters (enclosed by dotted lines) 1, 2, 3, 6 and 7, HBV_{LR} outperforms $DBv2_{LR}$ in terms of NSE for a majority of basins (ignoring open grey circle basins having similar performance). Conversely, for clusters 4 and 5, $DBv2_{LR}$ outperforms HBV_{LR}. Similar clusters can be observed when considering $GR4J_{LR}$ as the reference model (Fig. 7(c)). In this case, for basins represented by red dots, $GR4J_{LR}$ is superior to $DBv2_{LR}$. Furthermore, we identified a better-performing (difference in NSE = ± 0.05) model compared to the other two

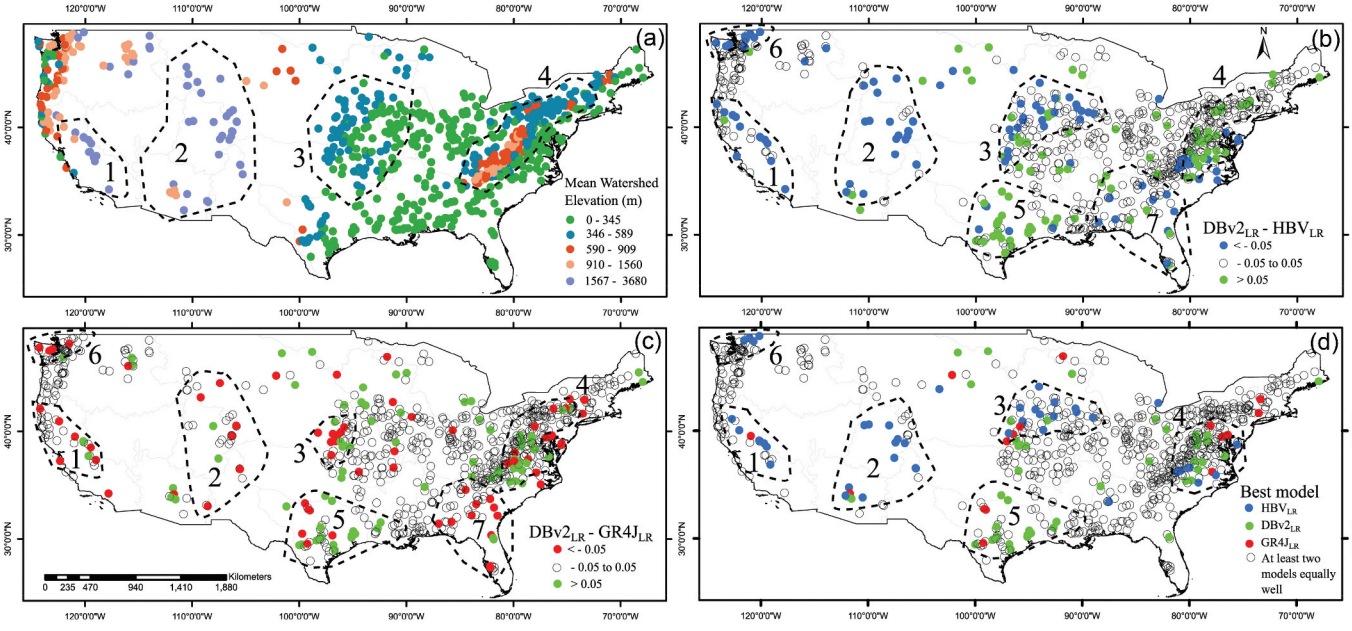


Figure 7. (a) Mean watershed elevation; (b) $DBv2_{LR}$ and HBV_{LR} performance comparison in terms of NSE; (c) $DBv2_{LR}$ and $GR4J_{LR}$ performance comparison in terms of NSE; (d) best model for the basin in terms of NSE.

models for a given basin. In Fig. 7(d), for a majority of basins in clusters 1, 2 and 3, HBV_{LR} outperforms both $DBv2_{LR}$ and $GR4J_{LR}$ (the difference in NSE is more than 0.05). On the other hand, for clusters 4 and 5, $DBv2_{LR}$ with a calibration-free runoff generation module outperforms HBV_{LR} and $GR4J_{LR}$ with calibrated runoff generation modules. For the open grey circle basins, at least two models are performing similarly. Interestingly, similar clusters (clusters 1 to 4) of basins are observed in the case of mean watershed elevation of the basins (Fig. 7(a)). The association of $DBv2_{LR}$ performance (considering $GR4J_{LR}$ and HBV_{LR} as benchmark models) with mean watershed elevation is evident and is shown in Fig. 8.

The $DBv2_{LR}$ performance with respect to HBV_{LR} and $GR4J_{LR}$ in terms of NSE shows a negative correlation ($r = -0.29$ and $r = -0.1$) with the mean watershed elevation of the basin. Similarly, in the case of the precipitation seasonality index (Fig. 8(b) and (c)), these correlation values are −0.17 and −0.09, respectively. This indicates that HBV_{LR} is superior to $DBv2_{LR}$ for basins with high mean watershed elevation and precipitation seasonality, while $GR4J_{LR}$ shows similar performance to $DBv2_{LR}$ in both cases with a correlation coefficient close to 0.1. The mean watershed elevation represents the spatial variation in rainfall, whereas precipitation seasonality is the temporal variation. Thus, HBV_{LR} is superior to $DBv2_{LR}$ and $GR4J_{LR}$ in the basins with high spatial and temporal variability, possibly due to the two extra free parameters (β and FC) in the runoff-generation module. A similar trend is observed in the case of snow percentage (Fig. 8(e) and (f)), where HBV_{LR} outperformed $DBv2_{LR}$ for basins with a high percentage of snow, while $GR4J_{LR}$ showed a similar performance to $DBv2_{LR}$. This is because snow parameters are calibrated for HBV_{LR}, thereby favouring its performance. Interestingly, the performance of $DBv2_{LR}$ with respect to HBV_{LR} and $GR4J_{LR}$ does not show a correlation with the dryness index of basin (Fig. 8(g) and (h)), which allows its applicability in a wide range of climates. In the case of percentage of deciduous forest (Fig. 8(i)), $DBv2_{LR}$ performance with respect to HBV_{LR} shows a positive correlation of 0.16. Moreover, for the mean annual PET, this value is 0.2. The proposed DBv2 runoff generation module is able to capture soil moisture dynamics better than HBV in basins with a high percentage of deciduous forest and mean annual PET. A possible explanation for this is that the structure of DBv2 is more favourable for basins with significant deciduous forests. However, the runoff generation mechanism of GR4J shows a very similar performance (Fig. 8(j) and (l)) to DBv2 in these basins.

3.3 Do we need a bucket?

In rainfall-runoff modelling practice, it is assumed that each catchment responds (streamflow response) uniquely to a given rainfall input due to the heterogeneity exhibited mainly by soil characteristics. This assumption results in the introduction of several free parameters into the hydrological model to account for dissimilarities in catchments. For example, typical bucket models such as HBV and PDM (Moore 2007) employ parameters to account for maximum soil storage capacity and distribution of storage. Estimation of these parameters is not only infeasible from field observations, but also their values obtained through model calibration with respect to discharge time series may not carry any physical meaning (e.g. Beven 1993, H. V. Gupta *et al.* 2006, Lan *et al.* 2020) – as Quintero *et al.* (2020) stated, "calibration kills understanding." At least, we need to explore the possibility of developing hydrological model structures that avoid the traditional calibration strategies. Gao *et al.* (2019) demonstrated that the runoff generation parameters of a bucket model can be obtained from topographic information. The underlying assumption behind their approach is that the parameters of a bucket model themselves are dependent on local climate (Gao *et al.* 2019, 2023). With the routing module only calibrated, they showed that their model performs as well as the fully calibrated HBV and TOPMODEL. Here, we made the same observation – the proposed model calibrated only for routing performs as well as the fully calibrated HBV and GR4J. However, it should also be emphasized here that our model brings a fresh perspective by doing away with the bucket structure, where maximum soil storage capacity is given a primary role. Instead, runoff generation is entirely governed by the *instantaneous dryness index* (Equation 6), a function of soil moisture storage. The proposed DBv2 model shifts the focus from a soil-centric to a climate-centric approach in runoff generation by excluding the explicit consideration of soil properties. The heterogeneity associated with soil properties in runoff generation is implicitly accounted for through climate inputs. Finally, our study and the study by Gao *et al.* (2019) highlight the importance of focusing on runoff routing, contradicting the earlier notion that runoff routing is less difficult than runoff generation modelling (V. K. Gupta *et al.* 1986, Beven 2011).

4 Conclusions

In recent times, Biswal (2016) proposed the calibration-free dynamic Budyko (DBv1) model to predict daily streamflow, which is based on the popular Budyko framework. Although this model showed acceptable performance in the case of ungauged basins, its low performance compared to a fully calibrated hydrological model was not investigated. In the present study, we analysed the DBv1 model for runoff generation and routing in predicting streamflow at basin outlet. The DBv1 estimates the instantaneous dryness index φ based on a power law-based decay function derived from recession flow analysis. In this study, we proposed an exponential decay function that improved the φ estimation and, ultimately, streamflow prediction. We compared the performance of the modified DBv1 (DBv2) runoff generation model with two well-established runoff generation models, HBV and GR4J. Our results showed that the proposed calibration-free DBv2 runoff generation model performed as well as calibrated HBV and GR4J runoff generation models in predicting daily streamflow for 975 CONUS basins.

It is assumed that soil plays an important role in runoff generation. Soil heterogeneity is accounted in terms of free parameters in the hydrological model. However, soil properties responsible for runoff generation are controlled by climate.

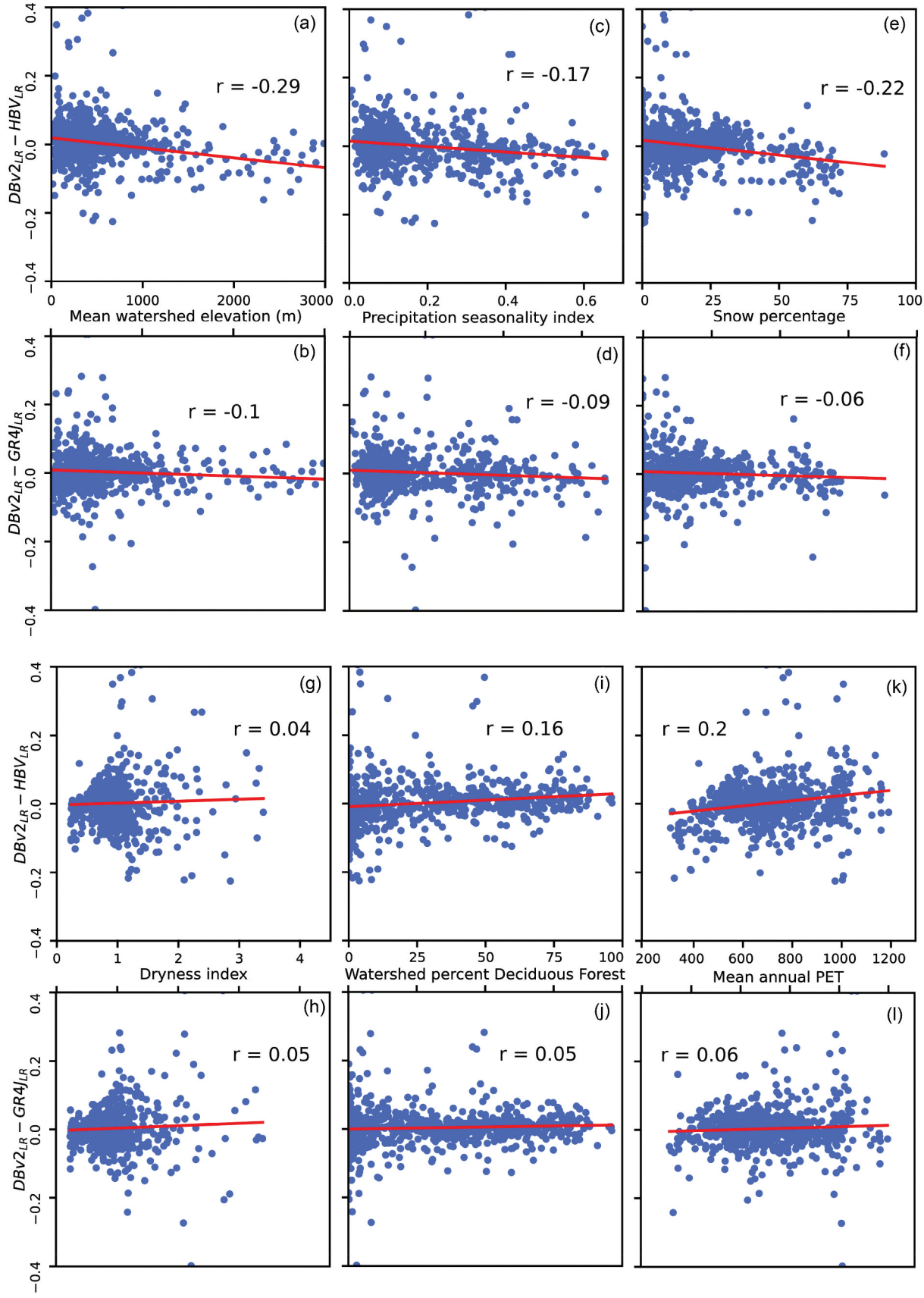


Figure 8. Correlation of $DBv2_{LR}$ performance with respect to HBV_{LR} and $GR4J_{LR}$ in terms of NSE with mean watershed elevation, precipitation seasonality index, snow percentage, dryness index, watershed percent deciduous forest and mean annual PET.

Thus, climate information is sufficient to partition the rainfall into streamflow and evapotranspiration, even at a daily time scale. This is well supported by our results as the calibration-free DBv2 runoff generation model performed as well as the fully calibrated HBV and GR4J runoff generation models. Thus, we should be focused on developing climate-centric hydrological models rather than soil-centric ones. Our study is the latest to highlight the need for a paradigm shift in rainfall-runoff modelling, with runoff generation simulation being performed with little or no calibration.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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